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StreamCatTools: An R package for working with StreamCat and LakeCat watershed data in R

Marc H. Weber[1], **Ryan A. Hill**[1], **Selia Markley**[2], **Travis Hudson**[3], **and Allen Brookes**[4]

1 Office of Water, United States Environmental Protection Agency **2** Oak Ridge Institute for Science and Education Fellow c/o United States Environmental Protection Agency **3** Oak Ridge Associated Universities Student Services Contractor c/o United States Environmental Protection Agency **4** United States Environmental Protection Agency Retired

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Summary

`StreamCatTools` provides an R interface to the StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) data and APIs for retrieving national scale watershed and lake-basin metrics. The package gives researchers and managers access to hundreds of catchment- and watershed-scale metrics without manual geospatial accumulation.

Statement of Need

Nationally consistent watershed data are essential for hydrology, water-quality assessment, and ecological modeling. The StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) datasets provide such data for every NHDPlusV21 (McKay et al., 2012) stream reach and lake basin in the conterminous United States (CONUS) and `StreamCatTools` makes these data accessible in R through a simple API wrapper for metric discovery and batched retrieval of data by regions such as NHDPlusV21 catchment (COMID), state, county, hydroregion, or CONUS, lowering the barrier to reproducible analysis and supporting the FAIR principles (Findable, Accessible, Interoperable, and Reusable) (Wilkinson et al., 2016).

State of the Field

R packages for watershed science usually cover one of three tasks: network navigation, observation retrieval, or general geospatial layers. `hydrogeofetch` (Blodgett, 2026) and related tools support stream-network traversal, `dataRetrieval` (DeCicco et al., 2026) supports hydrologic and water-quality observations, and many other packages provide raw geospatial data. The remaining gap is analysis-ready metrics already summarized to catchments and upstream watersheds and `StreamCatTools` provides programmatic access to these data in R, enabling researchers to retrieve covariates easily and reproducibly.

Package Overview

`StreamCatTools` is a lightweight R interface to the StreamCat API. The package allows for metadata discovery, metric lookup, and data retrieval using `httr2` (Wickham, 2026).

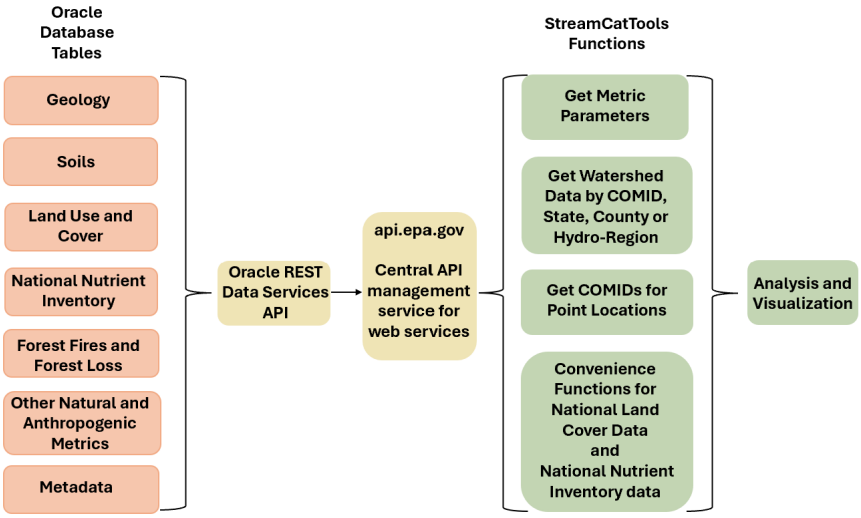


Figure 1: Diagram of the StreamCat and StreamCatTools framework. The backend Oracle database and REST web service are exposed through api.epa.gov to functions in StreamCatTools that simplify access and analysis of the data in R via the application programming interface (API). The example here shows the general categories of Oracle database tables in the StreamCat Oracle database.

The core functions retrieve variables at multiple AOIs: catchment (cat), watershed (ws), riparian catchment (catrp100), riparian watershed (wsrp100), and an other category for non-standard metrics Figure. They also support National Land Cover Database (NLCD) (U.S. Geological Survey, 2019) and National Nutrient Inventory (NNI) (Brehob et al., 2025) time-series products. The basic functionality and workflow is captured in 1.

You can install `StreamCatTools` from *CRAN* or GitHub:

```
install.packages("StreamCatTools")
```

```
install.packages("pak")
library(pak)
pkg_install("github::USEPA/StreamCatTools")
```

`StreamCatTools` is loaded into an **R** session:

```
library(StreamCatTools)
```

Users can inspect metric names and areas of interest before querying the API:

```
aois <- sc_get_params(param='aoi')
aois
```

```
#> [1] "cat"      "catrp100" "other"    "ws"       "wsrp100"
```

```
names <- sc_get_params(param='metric_names')
names[1:10]
```

```
#>  [1] "agkffact"      "al2o3"         "bankfulldepth" "bankfullwidth"
#>  [5] "bfi"           "canaldens"     "cao"           "cbnf"
#>  [9] "chem"          "clay"
```

Table 1: Example of variable information returned by the variable_info parameter in the sc_get_params function.

Metric	Short Description	Year	Units
agkffact[AOI]	Ag Soil Erodibility Kf Factor	NA	Unitless
bfi[AOI]	Base Flow Index	NA	Percent
huden[Year][AOI]	Mean Housing Density	2010	Count/Square Kilometer
inorgnwetdep[Year][AOI]	Mean Annual Precipitation-Weighted Nitrogen	2008	Kilogram/Hectare/Year
n_ags_[Year][AOI]	Nitrogen Agricultural Surplus	1987-2017	Kilograms
pcthighsev[Year][AOI]	Percent High Burn Severity Class For Year	1984-2018	Percent

Table 2: Example of metric names returned by the sc_get_metric_names function.

Category	Metric	AOI
Climate	bfi[AOI]	Cat, Ws
Deposition	inorgnwetdep[Year][AOI]	Cat, Ws
Deposition	nh4[Year][AOI]	Cat, Ws
Deposition	no3[Year][AOI]	Cat, Ws
Climate	precip8110[AOI]	Cat, Ws
Climate	precip9120[AOI]	Cat, Ws
Climate	precip[Year][AOI]	Cat, Ws
Deposition	sn[Year][AOI]	Cat, Ws
Climate	tmax8110[AOI]	Cat, Ws
Climate	tmax9120[AOI]	Cat, Ws

The same pattern applies to LakeCat metadata retrieval with `lc_get_params()`.

In StreamCat and LakeCat individual catchments (the local drainage area for a stream or lake) are aggregated to the watershed using either a weighted average or a count/sum depending on the metric (Hill et al., 2016). `sc_get_params()` and `lc_get_params()` return variable descriptions, units, years, and metric categories as shown in Table 1.

`sc_fullname()` and `lc_fullname()` return the full descriptive name for a given metric.:

```
sc_fullname(metric='pctgrs2019')
```

```
#> [1] "Grassland/Herbaceous Percentage 2019"
```

Users can also filter available metrics by year, category, dataset, and AOI using `sc_get_metric_names()` and `lc_get_metric_names()` (Table 2).

```
metrics <- sc_get_metric_names(category = c('Deposition','Climate'),
                               aoi=c('Cat','Ws'))
my_data <- head(metrics[,c('Category','Metric','AOI')],10)
```

The `sc_get_data()` and `lc_get_data()` functions allow users to extract catchment or watershed metrics by chosen regional unit - below we request two catchment and watershed

metrics (NLCD medium intensity developed land cover and density of dams) for three stream reaches:

```
df <- sc_get_data(metric='pcturbmd2019,damdens',
                  aoi='cat,ws',
                  comid='179,1337,1337420')
```

A parallel LakeCat request can be made by county or other spatial filter such as county FIPS codes for a state or county-level summary.:

```
df <- lc_get_data(metric='pctwdwet2006', aoi='ws', county='41003')
head(df)
```

```
#>      comid pctwdwet2006ws
#> 1 23769033      0.0000000
#> 2 23769029      0.1701645
#> 3 23769021      2.4697951
#> 4 23769003      0.0000000
#> 5 23768999      8.1185071
#> 6 23768983      0.0000000
```

Helper functions are available to discover the relevant FIPS codes for counties for data retrieval and simplify state- and county-level queries:

```
df <- sc_get_data(metric='pctwdwet2006',
                  aoi='ws', county='41003')
```

```
df <- sc_get_params(param='county') |>
  dplyr::filter(state == 'OR' & county_name == 'Benton County') |>
  dplyr::pull(fips)
```

```
# State-wide watershed queries are large; run the live call once, then
# cache to disk so subsequent knits read the cache instead of
# re-hitting (or stalling on) the API. Commit the .rds alongside paper.Rmd.
ct_cache <- "clay_agkffact_CT_ws.rds"
if (file.exists(ct_cache)) {
  df <- readRDS(ct_cache)
} else {
  df <- sc_get_data(metric = 'clay,agkffact', aoi = 'ws', state = 'CT')
  saveRDS(df, ct_cache)
}
head(df)
```

```
#>     comid agkffactws   clayws
#> 1 7702768     0.0177 6.100716
#> 2 7702200     0.0009 6.243493
#> 3 7702202     0.0011 6.368938
#> 4 7701112     0.0177 6.080299
#> 5 7701168     0.0054 6.092434
#> 6 7701214     0.0040 6.082292
```

`StreamCatTools` also provides convenience functions for the NLCD and NNI datasets through `sc_get_nlcd()`, `lc_get_nlcd()`, `sc_get_nni()`, and `lc_get_nni()`:

```
df <- sc_get_nlcd(comid='1337420', year='2019', aoi='ws')
```

```
df <- sc_get_nni(year='1987, 1990, 2005, 2017', aoi='cat,ws',
                 comid='179,1337,1337420')
```

Software design

`StreamCatTools` is designed as a lightweight wrapper around the StreamCat and LakeCat REST services. Requests are issued through `httr2`, with functions grouped into matching `sc_*` and `lc_*` families for stream and lake metrics. Data access functions accept metric names, spatial filters, and AOI syntax and return analysis-ready data. The package integrates with spatial workflows through `sf` and `hydrogeofetch`, and the lake-basin workflow uses DuckDB-backed geometry access for efficient retrieval.

Applications and Discussion

`StreamCatTools` can be used to link watershed metrics with spatial analyses and ecological models. For example, users can calculate county- or state-scale summaries of land cover or hydrologic stressors and then combine those outputs with `ggplot2` or `hydrogeofetch`-based mapping:

```
df <- sc_get_data(
  metric = "pctagslphigh2019",
  aoi = "ws",
  county = "41003"
) |>
  dplyr::summarise(
    mean_pctagslphigh2019ws = mean(
      pctagslphigh2019ws, na.rm = TRUE
    )
  )
df
```

```
#>   mean_pctagslphigh2019ws
#> 1                4.406472
```

```
library(hydrogeofetch)
library(ggplot2)
library(ggspatial)
library(StreamCatTools)

start_comid <- 23763517
nldi_feature <- list(
  featureSource = "comid",
  featureID = start_comid
)

flowline_nldi <- hydrogeofetch::navigate_nldi(
  nldi_feature,
  mode = "UT",
```

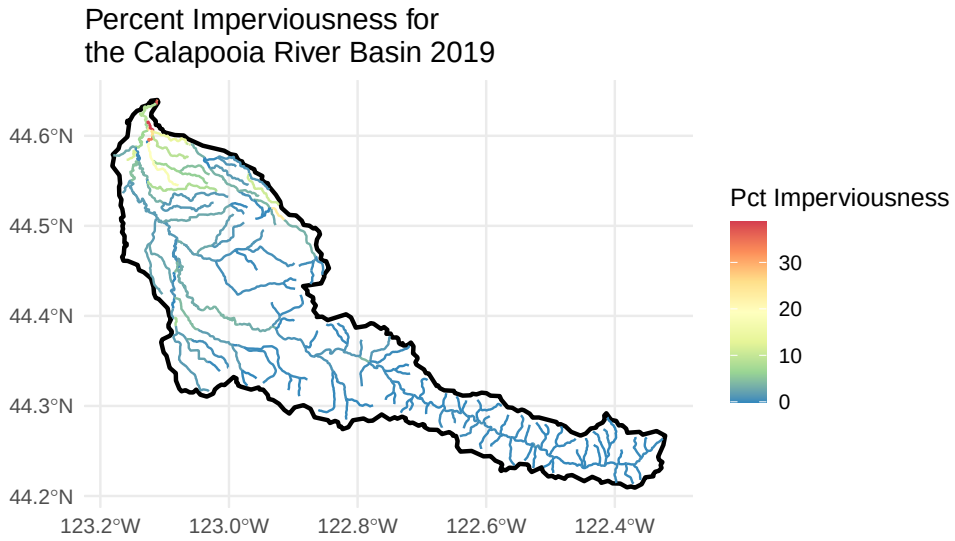


Figure 2: Map of NLCD percent imperviousness for each catchment for the Calapooia River watershed in Oregon.

```
  data_source = "flowlines",
  distance = 5000
)

df <- sc_get_data(
  metric = "pctimp2019",
  aoi = "cat",
  comid = flowline_nldi$UT_flowlines$nhdplus_comid
)

flowline_nldi <- flowline_nldi$UT_flowlines
flowline_nldi$PCTIMP2019 <- df$pctimp2019cat[
  match(flowline_nldi$nhdplus_comid, df$comid)
]

basin <- hydrogeofetch::get_nldi_basin(
  nldi_feature = nldi_feature
)
```

Figure 2 plots the NLCD percent imperviousness for the local drainage mapped to each stream reach and to the overall basin boundary.

Watersheds for lakes can also be retrieved using `lc_get_watershed()` to visualize lake-basin land-cover composition as in Figure 3.

```
library(ggplot2)
library(patchwork)
library(ggforce)
```

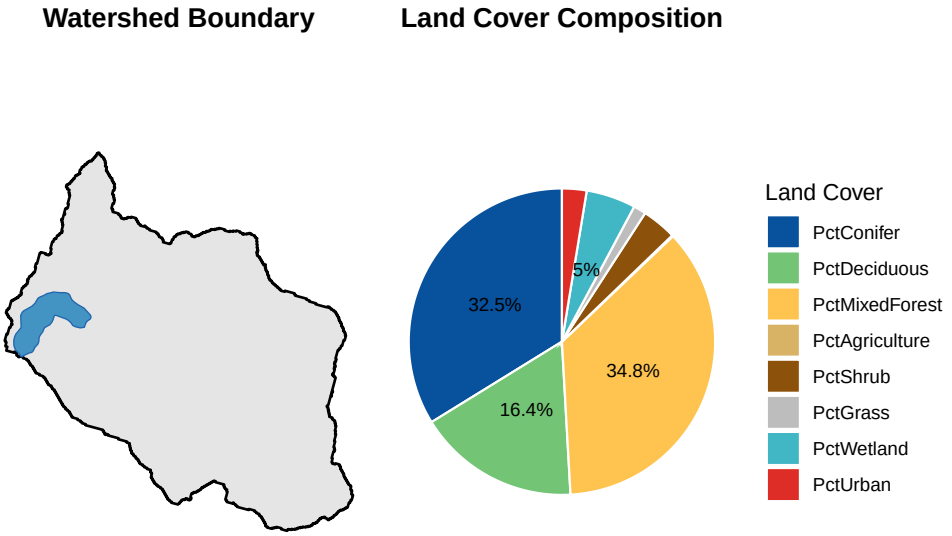


Figure 3: NLCD land cover proportions with an example lake watershed.

```
df <- lc_get_nlcd(comid='19334077', year='2019', aoi='ws')
lake <- hydrogeofetch::get_waterbodies(id = 19334077)
ws <- lc_get_watershed(comid = 19334077, huc2 = "01",
                       huc2_filter = "01",
                       threads = 2,retries = 5, verbose=FALSE,
                       progress=FALSE)
```

Functions for plotting NNI metrics and time-series exploration for nitrogen and phosphorus budgets are also included (Markley et al., 2026), enabling 4.

```
library(StreamCatTools)
library(ggplot2)
library(ggpattern)
com <- '22812041'
sc_plotnni(comid = com, include.nue = TRUE)
```

Future development includes plotting functions and additional metrics. A complementary package, `StreamCatR` is also being developed to allow users to process their own landscape metrics to hydrologic frameworks with network topology. This supports a broader objective of making national-scale watershed data more accessible and reproducible for management and research applications.

Research impact

Providing metric discovery, batched retrieval, and spatial integration in a single R interface, `StreamCatTools` facilitates use of this valuable CONUS scale data in monitoring, ecological assessment, and management decisions. `StreamCatTools` supports Clean Water

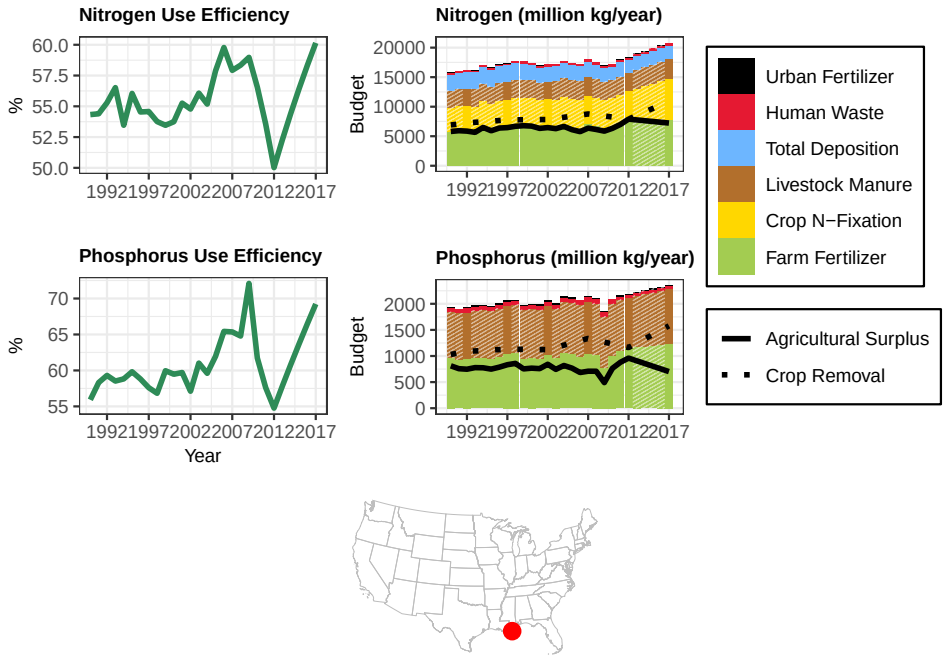


Figure 4: Annual time series of nitrogen and phosphorus budget data for the Mississippi-Atchafalaya River Basin.

Act authorities in state integrated reporting (i.e. Oregon Department of Environmental Quality (Oregon Department of Environmental Quality, n.d.) and Alabama Department of Environmental Management (Alabama Department of Environmental Management, 2025)), as well as species-distribution and aquatic-connectivity studies.

Acknowledgements

Examples of using StreamCat and LakeCat make extensive use of `hydrogeofetch` (Blodgett, 2026) and the functions for accessing the API are facilitated through use of `httr2` (Wickham, 2026). Figures were created using `ggplot2` (Wickham, 2016).

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AI Usage Disclosure

The R code for this R package was initially developed by Marc Weber and minimal generative AI was used for the most recent version of the package where several functions were refactored.

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